

SOP-WELD-1500: Welding and Cutting Safety and Procedure Qualification

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Section 1. Purpose and Scope

Governs safe practice for arc and gas welding/cutting, and the welding procedure specification (WPS) qualification required for pressure-boundary welds.

Section 2. Hot Work Interface

Clause 2.1 — All welding and cutting outside a permanent designated welding bay requires a Hot Work Permit per SOP-FIRE-610 in addition to this procedure.

Section 3. Procedure Qualification

Clause 3.1 — Any weld on a pressure-boundary component (piping, vessel, tank) shall be performed per a Welding Procedure Specification (WPS) qualified to ASME Section IX; field welds using an unqualified procedure are prohibited.

Clause 3.3 — The welder performing a pressure-boundary weld shall hold a current welder performance qualification (WPQ) specific to the WPS's process, position, and material group; a WPQ lapses after 6 months without performing that specific process, unless production welds in that process were performed and documented within the period.

Section 4. Inspection

Clause 4.1 — Pressure-boundary welds require the NDT method specified on the piping/vessel class datasheet (typically radiography or ultrasonic testing) before the weld is accepted; visual inspection alone is not sufficient for Class 1 or higher service.

Section 5. PPE

Clause 5.1 — Welding PPE (helmet with correct filter shade, flame-resistant clothing, gloves) is mandatory for the welder and any helper within 15 feet of arc welding operations.

Section 6. Records

Clause 6.1 — WPS qualification records, WPQ certifications, and weld NDT reports are filed to the Mechanical Welding File and retained for the life of the equipment welded.